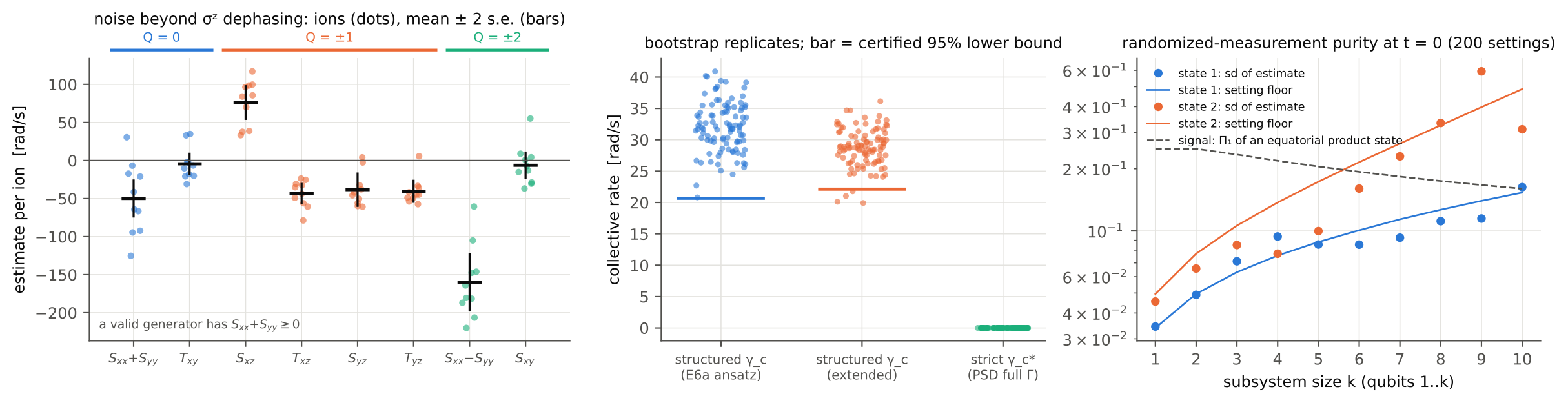
noise beyond σ² dephasing: ions (dots), mean ± 2 s.e. (bars)
Q = 0
Q = ±1
Q = ±2
estimate per ion [rad/s]
100
50
0
−50
−100
−150
−200
a valid generator has $S_{xx}+S_{yy} \geq 0$
$S_{xx}+S_{yy}$
T_{xy}
S_{xz}
T_{xz}
S_{yz}
T_{yz}
$S_{xx}-S_{yy}$
S_{xy}
bootstrap replicates; bar = certified 95% lower bound
collective rate [rad/s]
40
35
30
25
20
15
10
5
0
structured γ_c (E6a ansatz)
structured γ_c (extended)
strict γ_c* (PSD full Γ)
randomized-measurement purity at t = 0 (200 settings)
state 1: sd of estimate
state 1: setting floor
state 2: sd of estimate
state 2: setting floor
signal: Π₁ of an equatorial product state
6×10^{-1}
4×10^{-1}
3×10^{-1}
2×10^{-1}
10^{-1}
6×10^{-2}
4×10^{-2}
3×10^{-2}
1
2
3
4
5
6
7
8
9
10
subsystem size k (qubits 1..k)